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Factorisation

Factorising is expanding backwards. Take out the HCF, then count the terms — the number of terms tells you the method.

THE TOPIC ON ONE MAP

Factorisation

01 — Count the terms

 2 terms 3 terms 4 terms

02 — Two terms: the HCF

 $12pq - 18p$

"factorise fully"

03 — Two terms: DOTS

 $x^2 - 49$ $2x^2 - 18$

04 — Three terms: the CC Rule

 $x^2 + 2x - 24$ $2x^2 - x - 15$

hence solve

05 — Four terms: grouping

 $ax + ay + bx + by$

00 Count the Terms

Before anything: take out the **HCF**. Then count what is left — the count chooses the method.

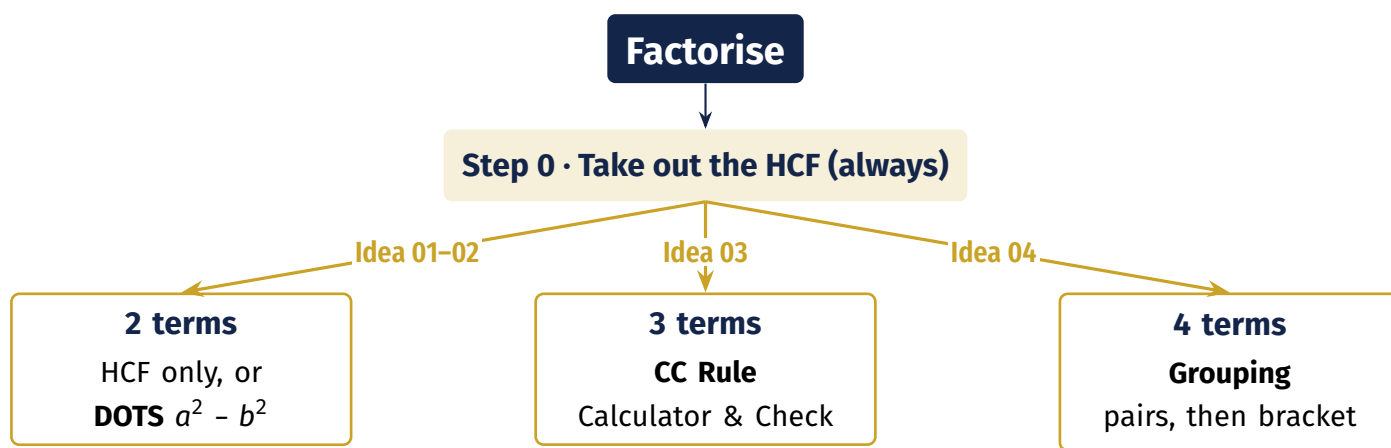
WHEN YOU SEE → YOU DO

“factorise fully” / “completely”

→ take out the biggest HCF first

count the terms

→ 2 · 3 · 4 terms → the method



BOTTOM LINE Factorising is the **first step** of algebraic fractions and of solving quadratics — and it comes on its own as a 1–3 mark question in almost every paper.

01 Two Terms: the HCF

Find what **every** term shares — the biggest number and the lowest power of each letter — and put it outside.

WHEN YOU SEE → YOU DO

“factorise fully” + two terms

→ biggest number, lowest power of each letter

“explain why ... is not correct”

→ something common is still inside the bracket

1

Numbers

HCF of the numbers: 16 and 20 → 4.

2

Letters

Each letter in **every** term, at its **lowest** power: a^2b^3 and a^3b → a^2b .

3

Divide to fill the bracket

Each term ÷ HCF. Then expand back in your head to check.

WORKED MODEL • FACTORISE FULLY

$$16a^2b^3 + 20a^3b$$

1 HCF = $4a^2b$

number • letters

2 $\frac{16a^2b^3}{4a^2b} = 4b^2$ $\frac{20a^3b}{4a^2b} = 5a$

divide each term

ANSWER $4a^2b(4b^2 + 5a)$

× **TRAP** “Fully” means the **biggest** HCF. $2x^2 + 6x = x(2x + 6)$ is not finished — 2 is still common: $2x(x + 3)$.

IDEA 01

Practise: Two Terms: the HCF

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Numbers and letters

CH 2 · TOPIC 4 2 marks

Factorise fully $15a^2b + 10ab^2$.

Answer _____

Q2 DR ESLAM Lowest power of each letter

CH 2 · TOPIC 4 2 marks

Factorise fully $12x^3y - 18x^2y^2$.

Lowest power of x: x^2 . Lowest power of y: y.

Answer _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 4 1 mark

1 Factorise $y^2 - 2y$

(1 mark)

Q4**PRACTICE**

Practice · Easy

CH 2 · TOPIC 4 1 mark

8 Factorise $y^2 + 27y$

(1 mark)

Q5**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 2 marks

13 Factorise completely $6y^2 - 9xy$

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks

7 Factorise fully $25a^4c^7d + 45a^9c^3h$

(2 marks)

PAST PAPERS**Q7****PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q9

CH 2 · TOPIC 4 5 marks

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$ $x =$

(Total for Question 9 is 5 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q10

CH 2 · TOPIC 4 7 marks10 (a) Simplify $(6m)^0$ (b) Solve $5y + 20 < 7y + 1$ (c) Factorise fully $15w^2x^5 + 25w^3x^2$ (d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$ **(Total for Question 10 is 7 marks)**

02 Two Terms: Difference of Two Squares

A minus sign between two perfect squares. It factorises in one line — and the HCF comes out first.

WHEN YOU SEE → YOU DO

two squares with a minus

→ $(a + b)(a - b)$

"...² - ..." after an HCF

→ HCF first, then DOTS

?

Is it DOTS?

1. A minus sign in the middle
2. First term is a square
3. Last term is a square

$$a^2 - b^2 = (a + b)(a - b)$$

$\sqrt{\text{first}}$ and $\sqrt{\text{last}}$, + and -

WORKED MODEL • HCF, THEN DOTS

$$50g^2 - 18$$

$$1 \quad 2(25g^2 - 9)$$

HCF first

$$2 \quad \sqrt{25g^2} = 5g \quad \sqrt{9} = 3$$

roots

ANSWER $2(5g + 3)(5g - 3)$

× **TRAP** $x^2 + 9$ does **not** factorise — DOTS needs a **minus**.

CALCULATOR CHECK Put $g = 2$ into both: $50(4) - 18 = 182$ and $2(13)(7) = 182$. Same number → correct.

IDEA 02

Practise: Two Terms: DOTS

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM HCF first, then DOTS

CH 2 · TOPIC 4 3 marks

Factorise fully (a) $5x^2 - 20$ (b) $12x^2 - 27$ (c) $2x^3 - 8x$

Q10 DR ESLAM Is it DOTS?

CH 2 · TOPIC 4 1 mark

Which of these can be factorised as a difference of two squares? Explain.

(a) $x^2 - 36$ (b) $x^2 + 36$ (c) $9y^2 - 1$

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 1 mark

6 Factorise $x^2 - 49$

(1 mark)

Q12**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks

1 Factorise fully $20x^2 - 5$

(2 marks)

Q13**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 4 marks

4 (a) Factorise $a^2 - b^2$

(1 mark)

(b) Hence, or otherwise, simplify fully $(x^2 + 4)^2 - (x^2 - 2)^2$

(3 marks)

Q14**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 1 mark

9 Factorise $4c^2 - 9d^2$

(1 mark)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q11

CH 2 · TOPIC 4 5 marks

11 (a) Factorise $9x^2 - 4y^2$

(2)

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

(Total for Question 11 is 5 marks)

Q16 PAST PAPER Edexcel 4MA1 · May 2023 · Paper 1H · Q14

CH 2 · TOPIC 4 3 marks

14 Factorise fully $50g^2 - 18$

(Total for Question 14 is 3 marks)

03 Three Terms: the CC Rule

Calculator & Check. The calculator gives you the roots; you turn the roots into brackets.

WHEN YOU SEE → YOU DO

“factorise $x^2 + bx + c$ ”

→ CC Rule: calculator roots → brackets

“(i) factorise (ii) hence solve”

→ each bracket = 0 — the calculator roots

a root like $\frac{3}{2}$

→ the bracket $(2x - 3)$

C1

Calculator

MODE → 5 → 3 (or MENU → Equation → Polynomial → 2). Enter a , b , c .
Read the two roots.

Root is a whole number $x = 5$ → $(x - 5)$ flip the sign

Root is a fraction $x = \frac{3}{2}$ → $(2x - 3)$ bottom goes with x

C2

Check

Multiply the first terms of your brackets. Do they make the x^2 term?
If not, put the missing number in front.

WORKED MODEL · THE CHECK STEP

$$4x^2 - 10x + 6$$

1 roots $x = \frac{3}{2}, x = 1$

calculator

2 $(2x - 3)(x - 1)$

roots → brackets

3 $2x \cdot x = 2x^2$, but we need $4x^2$

check

ANSWER $2(2x - 3)(x - 1)$

“HENCE SOLVE” Each bracket = 0 — the answers are the roots the calculator already gave you.

IDEA 03

Practise: Three Terms: the CC Rule

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Whole-number roots

CH 2 · TOPIC 4 2 marks

Factorise $x^2 + x - 12$.

Calculator roots: → flip the signs.

Answer _____

Q18 DR ESLAM Fraction roots

CH 2 · TOPIC 4 2 marks

Factorise (a) $2x^2 + 7x + 3$ (b) $6x^2 - x - 2$

A root of $\frac{2}{3}$ gives the bracket $(3x - 2)$.

Q19 **DR ESLAM** The Check step

CH 2 · TOPIC 4 2 marks

Factorise fully $4x^2 - 10x + 6$.

After the brackets, multiply the first terms. Is it $4x^2$?

Answer _____

SELECTED PRACTICE**Q20** **PRACTICE** Practice · Medium

CH 2 · TOPIC 4 2 marks

1 Factorise $x^2 + 3x - 10$

(2 marks)

Q21 **PRACTICE** Practice · Medium

CH 2 · TOPIC 4 2 marks

5 Factorise $x^2 + 3x - 4$

(2 marks)

Q22**PRACTICE**

Practice · Medium

CH 2 · TOPIC 4 2 marks16 Factorise $x^2 - 11x + 24$

(2 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 4 2 marks12 Factorise $3x^2 + 14x + 8$
(2 marks)

PAST PAPERS

Q24 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q7**CH 2 • TOPIC 4** **8 marks**

- 7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

 $x = \dots\dots\dots$

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

 $\dots\dots\dots$

- (c) (i) Factorise $y^2 - 2y - 48$

 $\dots\dots\dots$

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

 $\dots\dots\dots$ **(Total for Question 7 is 8 marks)**

Q25**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q9

CH 2 • TOPIC 4 5 marks

9 (a) Simplify $(2x^3y^5)^4$

.....

(b) (i) Factorise $x^2 + 5x - 36$

.....

(ii) Hence, solve $x^2 + 5x - 36 = 0$

.....

(Total for Question 9 is 5 marks)**Q26****PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q9

CH 2 • TOPIC 4 3 marks

9 (i) Factorise $x^2 + 2x - 24$

.....

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....

(Total for Question 9 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q8

CH 2 · TOPIC 4 4 marks8 (a) Simplify $(3x^2y)^0$

.....

(b) (i) Factorise $x^2 - 5x - 36$ (ii) Hence solve $x^2 - 5x - 36 = 0$

.....

(Total for Question 8 is 4 marks)**Q28****PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q9

CH 2 · TOPIC 4 5 marks9 (a) Simplify $x^4 \times x^5$

.....

(b) Simplify $(4y^2)^3$

.....

(c) Factorise $n^2 - 7n + 12$

.....

(Total for Question 9 is 5 marks)

Q29**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q6

CH 2 · TOPIC 4 5 marks

6 (a) Solve the inequality $5x - 7 \leq 2$

.....

(b) (i) Factorise $y^2 - 2y - 35$

.....

(ii) Hence, solve $y^2 - 2y - 35 = 0$

.....

(Total for Question 6 is 5 marks)

Q30 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 1HR · Q5**CH 2 · TOPIC 4** **5 marks**

5 (a) Expand $3c^3(c + 4)$

.....

(b) (i) Factorise $x^2 + 8x - 9$

.....

(ii) Hence, solve $x^2 + 8x - 9 = 0$

.....

(Total for Question 5 is 5 marks)

Q31 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 2H · Q11**CH 2 · TOPIC 4** **3 marks**

11 (i) Factorise $x^2 + 5x - 24$

.....

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....

(Total for Question 11 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q7

CH 2 • TOPIC 4 3 marks7 (a) Expand $4x(x - 5)$ (b) Factorise $y^2 - 9y + 20$ **(Total for Question 7 is 3 marks)**

Q33**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1H · Q6

CH 2 · TOPIC 4 · 8 marks6 (a) Simplify $(2c^4d^7)^3$

.....

(b) Find the value of $5y^0$ where $y > 0$

.....

(c) Factorise fully $16a^2b^3 + 20a^3b$

.....

(d) (i) Factorise $x^2 + 9x - 22$

.....

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....

(Total for Question 6 is 8 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2H • Q11**CH 2 • TOPIC 4** **3 marks**

11 (i) Factorise $x^2 + 9x - 22$

(ii) Hence, solve $x^2 + 9x - 22 = 0$

(Total for Question 11 is 3 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q7

CH 2 • TOPIC 4 9 marks7 (a) Simplify $g^9 \div g^2$

.....

(b) Expand $5k^2(k^3 + 4)$

.....

(c) (i) Factorise $x^2 - 2x - 63$

.....

(ii) Hence, solve $x^2 - 2x - 63 = 0$

.....

(d) Solve the inequality $7 - 2y < 3y - 12$

.....

(Total for Question 7 is 9 marks)

Q36**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q8

CH 2 · TOPIC 4 6 marks

8 (a) (i) Factorise $x^2 + 5x - 24$

(ii) Hence, solve $x^2 + 5x - 24 = 0$

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

(Total for Question 8 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q8

CH 2 · TOPIC 4 5 marks8 (a) Simplify $a^6 \times a^{10}$

.....

(b) Simplify $c^{30} \div c^{12}$

.....

(c) (i) Factorise $y^2 - 10y + 21$

.....

(ii) Hence, solve $y^2 - 10y + 21 = 0$

.....

(Total for Question 8 is 5 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q6

CH 2 · TOPIC 4 6 marks

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(b) (i) Factorise $y^2 - 11y + 30$

.....

(ii) Hence solve $y^2 - 11y + 30 = 0$

.....

(Total for Question 6 is 6 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q7

CH 2 · TOPIC 4 3 marks

7 (i) Factorise $x^2 + 2x - 48$

.....

(ii) Hence, solve $x^2 + 2x - 48 = 0$

.....

(Total for Question 7 is 3 marks)

04 Four Terms: Grouping

Two pairs. Factorise each pair — the same bracket appears twice. Take it out.

WHEN YOU SEE → YOU DO

four terms

→ group in pairs → the same bracket twice

1

Pair up

First two terms, last two terms.

2

HCF of each pair

The two brackets must come out **identical**. If not, swap the middle terms.

3

Take the bracket out

The repeated bracket is the common factor.

WORKED MODEL • GROUPING

$$xy + 3x + 2y + 6$$

1 $x(y + 3) + 2(y + 3)$

HCF of each pair

ANSWER $(y + 3)(x + 2)$

IDEA 04

Practise: Four Terms: Grouping

Questions written to practise exactly this idea.

Q40 DR ESLAM Grouping

CH 2 · TOPIC 4 2 marks

Factorise $xy + 3x + 2y + 6$.

Answer _____

Q41 DR ESLAM Grouping with minus signs

CH 2 · TOPIC 4 2 marks

Factorise $ax - ay + bx - by$.

Answer _____

05 Where You Will Use It

Factorising is the tool behind four later topics. Each one starts with “factorise”.

WHEN YOU SEE → YOU DO

“simplify fully” (a fraction)

→ factorise top and bottom, then cancel

“solve” a quadratic with clean roots

→ factorise

“prove ... multiple of”

→ factorise to prove

Algebraic fractions

Golden rule: factorise everything, then cancel.

Solving quadratics

Clean answers on the calculator → factorise.

Factorising

Algebraic proof

Factorise to prove: $12n = 2(6n)$ is even.

Rearranging formulas

Subject twice → take it out as a factor.

WORKED MODEL • ALGEBRAIC FRACTIONS

$$\frac{x^2 - 9}{x^2 + 5x + 6}$$

$$\text{1 } \frac{(x+3)(x-3)}{(x+2)(x+3)}$$

factorise top and bottom

$$\text{2 } \text{cancel } (x+3)$$

same bracket

ANSWER $\frac{x-3}{x+2}$

IDEA 05

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q42 DR ESLAM Algebraic fractions

CH 2 · TOPIC 4 3 marks

Simplify fully $\frac{x^2 + 4x}{x^2 - 16}$.

Factorise the top (HCF) and the bottom (DOTS), then cancel.

Answer _____

Q43 DR ESLAM Solving quadratics

CH 2 · TOPIC 4 2 marks

Solve $x^2 + x - 12 = 0$.

You factorised it in Idea 03: $(x + 4)(x - 3)$. Each bracket = 0.

$x =$ _____

Q44 DR ESLAM Proof: factorise to prove

CH 2 · TOPIC 4 3 marks

Show that $(n + 2)^2 - (n - 2)^2$ is a multiple of 8 for all integers n .

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 00 | Always first | Take out the HCF. Then count the terms. |
| 01 | 2 terms • HCF | Biggest number, lowest power of each letter.
“Fully” = biggest HCF. |
| 02 | 2 terms • DOTS | minus + two squares: $a^2 - b^2 = (a + b)(a - b)$ |
| 03 | 3 terms • CC Rule | Calculator → roots → brackets (flip sign; bottom goes with x) → Check the x^2 . |
| 04 | 4 terms | Group in pairs → same bracket → take it out. |
| 05 | Use it | Fractions (factorise first), quadratics, proof, rearranging. |

BOTTOM LINE HCF first → count the terms → 2: DOTS • 3: CC Rule • 4: grouping → expand back to check.

Answers

Factorisation • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TWO TERMS: THE HCF

- | | | |
|-----------|---|--|
| Q1 | Dr Eslam • Numbers and letters | $5ab(3a + 2b)$ |
| Q2 | Dr Eslam • Lowest power of each letter | $6x^2y(2x - 3y)$ |
| Q3 | Practice • Easy | $y(y - 2)$ |
| Q4 | Practice • Easy | $y(y + 27)$ |
| Q5 | Practice • Medium | $3y(2y - 3x)$ |
| Q6 | Practice • Hard | $5a^4c^3(5c^4d + 9a^5h)$ |
| Q7 | Edexcel 4MA1 • May 2021 • Paper 1H • Q9 | (a) $5a^4c^3(5c^4d + 9a^5h)$ (b) $x = -\frac{7}{4}$ |
| Q8 | Edexcel 4MA1 • November 2025 • Paper 1H • Q10 | (a) 1 (b) $y > 9.5$ (c) $5w^2x^2(3x^3 + 5w)$ (d) $3a^3c^4$ |

IDEA 02 • TWO TERMS: DOTS

- Q9** Dr Eslam • HCF first, then DOTS (a) $5(x + 2)(x - 2)$ (b) $3(2x + 3)(2x - 3)$ (c) $2x(x + 2)(x - 2)$
- Q10** Dr Eslam • Is it DOTS? (a) yes $(x + 6)(x - 6)$ (b) no — no minus (c) yes $(3y + 1)(3y - 1)$
 $(x + 7)(x - 7)$
- Q11** Practice • Medium $5(2x + 1)(2x - 1)$
- Q12** Practice • Hard (a) $(a + b)(a - b)$ (b) $(2x^2 + 2)(6) = 12(x^2 + 1)$
- Q13** Practice • Hard $(2c + 3d)(2c - 3d)$
- Q14** Practice • Hard (a) $(3x + 2y)(3x - 2y)$ (b) $\frac{5x - 6}{8x}$
- Q15** Edexcel 4MA1 • June 2022 • Paper 1H • Q11 $2(5g + 3)(5g - 3)$
- Q16** Edexcel 4MA1 • May 2023 • Paper 1H • Q14

IDEA 03 • THREE TERMS: THE CC RULE

- Q17** Dr Eslam • Whole-number roots $(x + 4)(x - 3)$
- Q18** Dr Eslam • Fraction roots (a) $(2x + 1)(x + 3)$ (b) $(3x - 2)(2x + 1)$
- Q19** Dr Eslam • The Check step $2(2x - 3)(x - 1)$
- Q20** Practice • Medium $(x + 5)(x - 2)$
- Q21** Practice • Medium $(x + 4)(x - 1)$
- Q22** Practice • Medium $(x - 3)(x - 8)$
- Q23** Practice • Hard $(3x + 2)(x + 4)$
- Q24** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q7 (a) $x = 6.5$ (b) $8m^2g^3(2m + 3g^2)$ (c) $(y - 8)(y + 6); y = 8, -6$
- Q25** Edexcel 4MA1 • January 2021 • Paper 1H • Q9 (a) $16x^{12}y^{20}$ (b) $(x + 9)(x - 4); x = -9, 4$
- Q26** Edexcel 4MA1 • May 2021 • Paper 2H • Q9 $(x + 6)(x - 4); x = -6, 4$
- Q27** Edexcel 4MA1 • November 2021 • Paper 2H • Q8 (a) 1 (b) $(x - 9)(x + 4); x = 9, -4$
- Q28** Edexcel 4MA1 • January 2022 • Paper 1HR • Q9 (a) x^9 (b) $64y^6$ (c) $(n - 3)(n - 4)$
- Q29** Edexcel 4MA1 • January 2022 • Paper 1H • Q6 (a) $x \leq \frac{9}{5}$ (b) $(y - 7)(y + 5); y = 7, -5$
- Q30** Edexcel 4MA1 • June 2022 • Paper 1HR • Q5 (a) $3c^4 + 12c^3$ (b) $(x + 9)(x - 1); x = -9, 1$
- Q31** Edexcel 4MA1 • June 2022 • Paper 2H • Q11 $(x + 8)(x - 3); x = -8, 3$
- Q32** Edexcel 4MA1 • May 2023 • Paper 1HR • Q7 (a) $4x^2 - 20x$ (b) $(y - 4)(y - 5)$
- Q33** Edexcel 4MA1 • May 2023 • Paper 1H • Q6 (a) $8c^{12}d^{21}$ (b) 5 (c) $4a^2b(4b^2 + 5a)$ (d) $(x + 11)(x - 2); x = -11, 2$
- Q34** Edexcel 4MA1 • June 2024 • Paper 1H • Q6 (a) $(x + 11)(x - 2); x = -11, 2$

IDEA 04 • FOUR TERMS: GROUPING**Q40** Dr Eslam • Grouping

$$(x + 2)(y + 3)$$

Q41 Dr Eslam • Grouping with minus signs

$$(a + b)(x - y)$$

IDEA 05 • WHERE YOU WILL USE IT**Q42** Dr Eslam • Algebraic fractions

$$\frac{x}{x - 4}$$

Q43 Dr Eslam • Solving quadratics

$$x = -4 \text{ or } x = 3$$

Q44 Dr Eslam • Proof: factorise to prove

$$(n^2 + 4n + 4) - (n^2 - 4n + 4) = 8n, \text{ and } 8n = 8 \times n$$